

ARUN KUMAR SEKAR

Full-Stack Software Engineer

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Profile

Full-stack Software Engineer (MSc Computer Science, Distinction) who builds and operates production systems end-to-end — application development, AWS infrastructure, CI/CD, and security. Currently leads a cross-functional engineering team at a fast-growing startup, owning a core product within a multi-tenant SaaS suite used by hundreds of daily users. Published human-computer interaction researcher. Comfortable owning delivery from architecture through deployment and reliability, and turning business requirements into maintainable, scalable systems.

Work Experience

AR Experts Ltd

Manchester, United Kingdom (Remote)

Software Developer

April 2025 – Present

- Lead a cross-functional team of 3 web and mobile engineers delivering the BizAlign suite — 4 enterprise products (8 iOS / Android apps) — and own one core product end-to-end.
- Deliver a Workforce app used by around 600 drivers and a multi-tenant HR platform serving 3 client organisations, alongside Fleet and Telematics products.
- Replaced a manual vehicle inspection-and-signature chain (22-angle photo capture → document assembly → external DocuSign → per-driver Dropbox filing) with a single in-app flow: assistants capture inspections in-app, documents auto-generate and route for in-app signature across web and mobile — cutting turnaround from days to minutes and manual effort by around 60%.
- Own architecture and delivery of full-stack MERN applications with TypeScript throughout, prioritising modular, maintainable systems.
- Deployed and managed Dockerised ML model containers on EC2, reducing manual image-validation workload by around 75%.
- Build and maintain CI/CD pipelines (GitHub Actions) and manage AWS infrastructure (EC2 on Amazon Linux, S3); ship feature updates and fixes on a roughly bi-weekly cadence across all 4 apps.
- Lead code review, pull-request approval, App Store / Play Store deployment, and the team's DevOps and security practices.

Associate Software Developer

Nov 2024 – April 2025

- Delivered a multi-tenant ERP system handling 20,000–30,000 daily API calls and 10,000+ monthly uploads.
- Designed RESTful APIs and integrated third-party services for real-time data processing.
- Architected secure internal access with WireGuard VPN; administered Ubuntu and Amazon Linux servers (deployments, access control, monitoring).
- Set up development, staging, and production environments and ensured GDPR compliance for App Store and Play Store releases.

Northumbria University / Northumbrian Water Ltd

Newcastle, United Kingdom

Technical Lead — Advanced Practice Project

Jan 2024 – May 2024

- Led a team building a 3D flood-simulation system in Unreal Engine for a regulated public utility, integrating models from Blender, 3ds Max, and SketchUp.
- Defined system design and core mechanics and delivered the project on schedule.

Digitally Inspired Media Pvt Ltd

Chennai, India

Junior Graphic Designer

Sep 2021 – Jan 2023

- Produced 200+ creatives across 10+ brands; acted as interim Design Lead, reviewing and approving creative output.

Education

Northumbria University

Newcastle, United Kingdom

MSc Computer Science with Advanced Practice — **Distinction**

Jan 2023 – Jan 2025

St. Joseph's College of Engineering

Chennai, India

B.E. Electrical & Electronics Engineering, GPA: 8.43

Aug 2017 – April 2021

Core Skills

Programming: TypeScript, JavaScript, Python, Go, C#, Java, SQL

Full-Stack & Mobile: React, Node.js, Express, MongoDB (MERN); React Native, Expo; App Store / Play Store deployment

Cloud & DevOps: AWS (EC2, S3), Docker, CI/CD (GitHub Actions), Linux (Ubuntu, Amazon Linux), WireGuard VPN

Architecture: REST APIs, multi-tenant systems, microservices, scalable & secure system design

Data & Monitoring: Dashboards, data visualisation, Grafana, Loki, logging & anomaly detection

Practices: Agile / Scrum, code review, GDPR compliance, secure infrastructure

Also: Unreal Engine, Blender; IoT / embedded systems (ESP32, Arduino)

Publication & Selected Projects

- **Immersive Data Visualisation in VR** — Published in the *International Journal of Human-Computer Interaction* (2025). Built a VR data-visualisation system blending artistic patterns with charts; user studies demonstrated improved engagement and usability.
- **Autonomous Table Robot** — ESP32 embedded build with motor control, IR / ultrasonic sensing, IMU, and OLED display (personal IoT project).
- **Tooneland Event Platform** — Dynamic event-booking platform with user management and booking workflows.

Open Source

- **dbsnap** (Go) — Production-grade MongoDB backup / restore CLI; scheduled, compressed, and OS-keychain-secured. [github](#) ↗
- **quant_lab** (Python) — AI-powered algorithmic trading research platform; backtesting engine, strategy screener, and walk-forward validation. [github](#) ↗
- **expo-camera-modal** (React Native) — Published open-source full-screen camera modal for Expo; plug-and-play drop-in component. [github](#) ↗
- **git-shortcuts** (Shell) — Natural-language Git toolkit for Zsh and PowerShell. [github](#) ↗

Leadership

Vice Chairman & Treasurer, IEEE Robotics Society · Student Representative, MSc Computer Science, Northumbria University